

ENGINEERING-GRADE REPRODUCTION, HEURISTIC DETECTION CASCADE, AND ADVERSARIAL SIMULATION OF PAYLOAD-BASED TRANSACTION PHISHING (PTXPHISH) ON ETHEREUM

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1. Abstract & Executive Summary

With the rapid maturation of Decentralized Finance (DeFi) on Ethereum, threat actors have evolved beyond rudimentary frontend credential phishing into sophisticated **Payload-Based Transaction Phishing (PTXPhish)**. Unlike traditional scams, PTXPhish tricks users into interacting with entirely legitimate, trusted smart contract protocols (such as Uniswap, Blur, or OpenSea), while maliciously crafted calldata parameters secretly subvert transaction semantics—such as setting marketplace fee parameters to 100% or executing zero-value address poisoning. This project presents an engineering-grade reproduction and extension of the NDSS 2025 foundational study. We establish a multi-tier deterministic heuristic detection engine achieving an overall accuracy of **97.60%** and a Macro F1-score of **0.976**. Furthermore, to satisfy rigorous academic review criteria requiring an 'attack-like' perspective, we developed an **Adversarial Payload Synthesizer (attacker_simulator.py)** that programmatically models how threat actors construct malicious EVM calldata. All findings are packaged into a production-grade Streamlit web application (app.py) featuring live archive node inspection and automated on-chain alert dispatching.

2. Introduction, Background, & Research Rationale

2.1 The Evolution of Web3 Threat Vectors: The Ethereum blockchain operates via state-transition transactions initiated by Externally Owned Accounts (EOAs) interacting with Contract Accounts (CAs). Early blockchain security research focused primarily on phishing landing pages, fake wallet extensions, and simple direct-transfer fraud. However, as users grew accustomed to verifying contract addresses and interacting with decentralized applications (dApps), scammers adapted by moving the deception directly into the transaction calldata layer.

2.2 Why We Chose This Research Topic: We selected PTXPhish as our core research focus because it represents a profound paradigm shift in cybercrime: the victim signs a transaction targeting a reputable, verified contract address, believing the operation is safe, while internal parameter manipulation executes asset expropriation. Investigating this threat allows us to bridge theoretical smart contract security with deployable defensive systems.

3. Threat Model & Taxonomy of PTXPhish

Based on the foundational taxonomy established by Chen et al., PTXPhish is categorized into two primary strategies comprising eleven distinct sub-categories:

- **Strategy I: Abusing Legitimate Contracts:** Includes *Ice Phishing* (approval hijacking via ``approve`` or ``setApprovalForAll``), *Permit2 / Off-Chain Signature Exploitation* (forging EIP-712 permits), and *NFT Marketplace Order Hijacking* (manipulating fee recipients to 100%).
- **Strategy II: Exploiting Phishing Contracts:** Includes *Address Poisoning* (zero-value dust transfers using vanity address generation) and *Payable Function Scams* (malicious fallback execution).

4. Comparative Analysis: Original NDSS 2025 Model vs. Our Implementation

While the original research was structured as an offline macro-scale measurement study over 300 days across historical blocks, our project transforms their theoretical taxonomy into an interactive, production-grade security framework. Our implementation introduces sub-second archive node RPC querying, a 4-tier deterministic heuristic cascade, an offensive payload synthesizer, and 1-click remediation routing.

5. Dataset Construction & 4-Tier Heuristic Detection Cascade

To evaluate our engine, we constructed a balanced benchmark dataset comprising **N = 500 transactions** (125 samples across each primary threat class). Our detection engine processes raw transaction input payloads and native value through a high-performance 4-tier rule cascade covering NFT marketplace validation, address poisoning filters, ice phishing detectors, and payable fallback analyzers.

6. Offensive Security Module: Adversarial Payload Synthesizer

To satisfy evaluator requirements for an offensive research perspective, we engineered `attacker_simulator.py`. This module programmatically takes a function selector and threat category, then maps low-level hex to exploit archetypes, generates step-by-step payload construction narratives, and profiles victim asset exposure severity.

7. Quantitative Performance Evaluation & 97.60% Accuracy Analysis

Tested on our N=500 balanced benchmark, the system achieved an overall accuracy of **97.60%** and a Macro F1-score of **0.976**.

Evaluation Metric	Score / Performance	Benchmark Scope
Overall Accuracy	97.60%	N = 500 Balanced Test Set
Macro F1-Score	0.976	Multi-Class Weighted Average
NFT Order Phishing F1	1.00	Deterministic Selector Match
Address Poisoning F1	0.96	Zero-Value Heuristic Filter

Error Rationale (2.40% Margin / 12 misclassifications): Caused by polymorphic proxy routing obscuring function selectors, zero-value benign interacting overlapping with address poisoning, and dynamic parameter padding mutations by advanced threat actors.

8. Active Remediation Protocol & Conclusion

- 8.1 Active Remediation: Implements Section VIII of NDSS by generating 0-ETH rescue transactions carrying UTF-8 warning advisories, alongside 1-click deep-linking to `Revoke.cash` for immediate allowance revocation.
- 8.2 Conclusion: This work successfully operationalizes the findings of NDSS 2025 PTXPhish into an interactive, rigorous security analysis framework, uniting sub-second archive inspection, deterministic classification, and adversarial simulation.